
Gyroscopes stopped being this clunky more than two decades ago

some set path, and this data can be used to calculate, for instance, the speed you're walking/driving/cycling at.

Most often, technology devices use both accelerometer and gyroscope (A-G) combinations to get proper orientation and acceleration/speed data. The Nintendo Wii when it was first launched used exactly this method to allow you to play motion-based gaming, and forever changed the way we looked at consoles and games.

This is the common, personal way in which you are exposed to A-Gs, but they have their uses in many more facets of life.

Auto-pilot systems, which we mentioned before, depend heavily on A-Gs integrated with GPS to function. Using the accelerometer (coupled with many more sensors), the aircraft is able to judge its own direction of motion, and when you add in the gyroscopes, it is able to also control angle of tilt, etc. All of this data when fed into a pre-programmed system is able to control the throttle and flaps system of the aircraft to ensure that it keeps flying in the right direction, and as steadily as possible. Of course we're talking MEMS-based sensors here, not actual old-styled gyroscopes and other sensors. For those who don't already know, MEMS stands for Micro-Electro-Mechanical Systems, which are basically micrometer versions of sensors such as gyroscopes, but made in silicon and also containing a microprocessor to receive and transmit the readings.

What applies to aircraft also applies to other modes of transport such as your car. A-Gs are able to work together in your car, along with other sensors, such as traction control and the Anti-lock Braking System (ABS) to ensure the safest possible ride. Many of these sensors work together to form the ESC/ESP (Electronic Stability Control/Program) feature, which is available on most mid- to high-end cars. For example, this helps the car systems adjust the steering sensitivity when at higher speeds, or by reducing engine power when going around a corner to prevent skidding, or detecting that you didn't press lightly on the brake pedal, but stamped on it, and thus perhaps want to stop faster.

That's not all, if you bought a digital camera in recent time which features image stabilisation, this too is achieved through A-G systems, to offer you clear, jitter-free images. It's also used in everything from ships to cargo trucks, gaming consoles to the international space station, the Hubble telescope, the Segway, hard drives, and of course all smart phones, tablets and the like.

Sensors galore

Sensors have permeated our lives to such an extent that we just don't acknowledge their existence anymore. In fact, anything without sufficient sensors is actually considered dumb these days. We've all read and heard about accelerometers and gyroscopes, but for a better understanding of how they function, head over to http://dgit.in/CSJuly4

Medical

Even the medical world is changing due to sensors becoming cheaper and more abundant. Today, if you're worried about blood sugar or blood pressure, you can go to pretty much any medical store and pick up a home testing kit. These battery operated kits are able to easily identify your pressure or the amount of sugar in your blood and make it easier for those with health problems to monitor themselves, rather than rush to the hospital or to the doctor for a quick checkup.

Even X-ray scans, which used to need film to capture an image, and also resulted in that film being the only copy of the scan, have gone digital, using sensors to pick up the image – much like a digital camera does. The result is that you have your X-ray stored digitally, which can be "printed" multiple times, and sent across the Internet to far away experts even. Since chemicals aren't used (in the film) the resolution and clarity can be improved, and the ability to process the image after capturing it – like Photoshoping a bad image.

In medicine, prevention is always better than cure, and catching a disease or medical condition early is helping you live longer. Everything from MRI (Magnetic Resonance Imaging) to pacemaker monitors are sensor driven. With wireless power coming of age, research is underway to develop microscopic sensors than can be embedded under your skin, and powered by external handheld receiver devices when a reading is needed. This would make it easy for doctors to keep a watch on your stats without invasive procedures such as injections and drawing blood.

Even exercise and sports regimens need to be watched, and sensors are already easily available to tell you how much you have walked, how many calories you have burnt, etc. People who undergo bypass surgeries also need to adjust medicine doses as per their weight, which can fluctuate, and sensors can also be used to calculate the modified doses.

Apart from this, almost every medical procedure you've ever undergone – from a simple X-ray to a sonography, or checking your weight on a digital scale to being hooked up to machines in an intensive care unit – has sensors at the heart of the diagnostics. If you have a greater life expectancy now than ever before, it's all thanks to tiny little sensors doing their job well and on time.

Sensing the future

Sensors are going to be huge in the future, and always invisible. Just as we need our